

NAME: CH. SRI LAKSHMI

REG NO: 192311342

COURSE CODE: CSA1511

COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSESSMENT TOOL-14

1. HDFS vs Traditional File Systems

“HDFS is designed to store very large data across multiple machines, while traditional file systems usually store data on a single system. HDFS also provides replication and fault tolerance.”

2. Importance of Replication

“Replication keeps multiple copies of the same data on different nodes. If one node fails, another copy can be used, preventing data loss.”

3. MapReduce

“The Map phase processes input data and produces key-value pairs. The Reduce phase groups and processes those values to generate the final output.”

4. Role of YARN

“YARN manages the resources of a Hadoop cluster, such as CPU and memory. It also schedules and monitors different jobs running in the cluster.”

5. NAS vs SAN

“NAS provides file-level storage over a network, while SAN provides block-level storage. SAN is generally used when high performance is required.”

6. Purpose of ETL

“ETL means Extract, Transform, and Load. It collects data from different sources, cleans and transforms it, and loads it into a target system.”

7. Apache NiFi

“Apache NiFi helps automate the movement of data between different systems. It provides visual data flows, monitoring, and data transformation.”

8. Hadoop Failure Points

“Common failures include DataNode, NameNode, network, and resource failures. Replication, NameNode High Availability, monitoring, and recovery mechanisms help handle them.”

9. Secure Data Ingestion

“I would use encryption, authentication, and authorization to protect the data during ingestion. I would also use access controls and monitoring to detect security issues.”

10. Scalable Hadoop System

“Hadoop divides large data and processing tasks across multiple machines. As data increases, we can add more machines to handle the workload efficiently.”

